

RODOLFO G. BLANCO-RODRÍGUEZ

Postdoctoral Fellow · Department of Mathematics and Statistical Science · University of Idaho

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RESEARCH INTERESTS

Mathematical and computational modeling of complex biological and physical systems across spatial and temporal scales. My work combines mechanistic models—ordinary and partial differential equations, stochastic and agent-based models, molecular dynamics, and Monte Carlo simulation—with statistical inference and machine learning, including model calibration, parameter identification, clustering, and topological data analysis. Current applications include within-host viral dynamics and immune responses, the link between within-host and between-host transmission, epidemics on contact networks, and antimicrobial therapy design; earlier work addressed stress distribution and flow in granular materials.

Keywords: mathematical biology · infectious disease modeling · immunology · model calibration · agent-based models · machine learning · granular matter

ACADEMIC APPOINTMENTS

Postdoctoral Fellow

Nov 2022 – present

Department of Mathematics and Statistical Science, University of Idaho, Moscow, ID, USA

Supervisor: Prof. Esteban A. Hernández-Vargas

Postdoctoral Fellow

Sep 2020 – Aug 2022

Institute of Mathematics, Universidad Nacional Autónoma de México (UNAM), Juriquilla, Querétaro, Mexico

Supervisor: Prof. Esteban A. Hernández-Vargas

EDUCATION

Ph.D. in Applied Physics

Sep 2015 – Oct 2019

Department of Applied Physics, Cinvestav, Mérida, Yucatán, Mexico

Dissertation: *Numerical study of stress distribution in confined and flowing granular matter*

Advisor: Prof. Gabriel Pérez-Ángel

M.Sc. in Applied Physics

Sep 2013 – Aug 2015

Department of Applied Physics, Cinvestav, Mérida, Yucatán, Mexico

Thesis: *Stress distribution in two-dimensional silos*

Advisor: Prof. Gabriel Pérez-Ángel

B.Eng. in Engineering Physics

Sep 2006 – Jun 2013

Faculty of Engineering, Universidad Autónoma de Yucatán, Mérida, Yucatán, Mexico

Thesis: *Enhancement of thermal interface materials with carbon nanofibers*

Advisor: Prof. Rubén A. Medina-Esquível

Research stays:

Jan – Feb 2020 Forschungszentrum Jülich, Germany (host: Dr. Thorsten Auth). *Simulation of red blood cell membrane disruption by merozoites.*

Jun – Aug 2019 Universidad de Navarra, Pamplona, Spain (hosts: Prof. Diego Maza Ozcoidi and Prof. Raúl Cruz Hidalgo). *Effect of hopper angle on granular flow*; led to publication J2.

Blanco-Rodríguez, R. in bold. Total: 11 journal articles, 2 peer-reviewed conference papers.

Peer-reviewed journal articles

- J11. **Blanco-Rodríguez, R.**, Chi, L. A., & Hernández-Vargas, E. A. (2026). Multicompartment modeling of influenza A replication along the murine respiratory tract. *Frontiers in Virology*, 6, 1772993. doi:10.3389/fviro.2026.1772993
- J10. **Blanco-Rodríguez, R.**, Miura, T. A., & Hernández-Vargas, E. A. (2025). CrossLabFit: A novel framework for integrating qualitative and quantitative data across multiple labs for model calibration. *PLOS Computational Biology*, 21(11), e1013704. doi:10.1371/journal.pcbi.1013704
- J9. Anderson, A., Kinahan, M., **Blanco-Rodríguez, R.**, Gonzalez, A. H., Udekwu, K., & Hernández-Vargas, E. A. (2025). Computational platform for streamlining the success of sequential antibiotic therapy. *npj Antimicrobials and Resistance*, 3, 90. doi:10.1038/s44259-025-00160-w
- J8. **Blanco-Rodríguez, R.**, Anderson, A., & Hernández-Vargas, E. A. (2025). Linking within-host immune dynamics to between-host transmission and reinfection risk. *Journal of Theoretical Biology*, 112210. doi:10.1016/j.jtbi.2025.112210
- J7. **Blanco-Rodríguez, R.**, Tetteh, J. N., & Hernández-Vargas, E. A. (2024). Assessing the impacts of vaccination and viral evolution in contact networks. *Scientific Reports*, 14, 15753. doi:10.1038/s41598-024-66070-5
- J6. Chi-Uluac, L. A., Asgharpour, S., **Blanco-Rodríguez, R.**, & Martínez-Archundia, M. (2023). Atomistic molecular insights into angiotensin-(1-7) interpeptide interactions. *Journal of Chemical Information and Modeling*. doi:10.1021/acs.jcim.3c00478
- J5. **Blanco-Rodríguez, R.**, Ordoñez-Jiménez, F., Almocera, A. E. S., Chinney-Herrera, G., & Hernández-Vargas, E. A. (2023). Topological data analysis of antibody dynamics of severe and non-severe patients with COVID-19. *Mathematical Biosciences*, 361, 109011. doi:10.1016/j.mbs.2023.109011
- J4. Rodríguez-Martínez, E. A., Rivas, C. U., Ayala, M. E., **Blanco-Rodríguez, R.**, Juárez, N., Hernández-Vargas, E. A., & Aragón, A. (2023). A new computational approach, based on image trajectories, to identify the subjacent heterogeneity of sperm to the effects of ketanserin. *Cytometry Part A*. doi:10.1002/cyto.a.24732
- J3. **Blanco-Rodríguez, R.**, Du, X., & Hernández-Vargas, E. A. (2021). Computational simulations to dissect the cell immune response dynamics for severe and critical cases of SARS-CoV-2 infection. *Computer Methods and Programs in Biomedicine*, 211, 106412. doi:10.1016/j.cmpb.2021.106412
- J2. **Blanco-Rodríguez, R.**, Hidalgo, R. C., Pérez-Ángel, G., & Maza, D. (2021). Critical numerical analysis of quasi-two-dimensional silo-hopper discharging. *Granular Matter*, 23(4), 1–10. doi:10.1007/s10035-021-01159-6
- J1. **Blanco-Rodríguez, R.**, & Pérez-Ángel, G. (2018). Stress distribution in two-dimensional silos. *Physical Review E*, 97(1), 012903. doi:10.1103/PhysRevE.97.012903

Peer-reviewed conference papers

- C2. **Blanco-Rodríguez, R.**, Humphrey IV, A., Anderson, A., & Hernández-Vargas, E. A. (2025). Parameter identification for PDEs via qualitative constraints. *IFAC-PapersOnLine*, 59(8), 72–77 (5th IFAC Workshop on Control of Systems Governed by PDEs, Beijing, China). doi:10.1016/j.ifacol.2025.08.069
- C1. **Blanco-Rodríguez, R.**, & Hernández-Vargas, E. A. (2024). Calibration of stochastic dynamic networks: An application to epidemics. In *World Automation Congress* (pp. 46–57; WAC 2024, Cancún, Mexico). Springer Nature Switzerland. doi:10.1007/978-3-032-10150-1_5

Manuscripts in preparation

- **Blanco-Rodríguez, R.**, et al. Interferon-mediated antiviral protection limits influenza spread: A spatial agent-based model. *Expected submission: Fall 2026*.
- **Blanco-Rodríguez, R.**, et al. A stochastic germinal center model in antigenic space for cross-variant antibody responses to SARS-CoV-2 vaccination. *Expected submission: Fall 2026*.

GRANTS, FELLOWSHIPS, AND AWARDS

Fellowships

2020 – 2022	Postdoctoral Fellowship, Institute of Mathematics, UNAM, Mexico (MXN 27,000/month, 2 years)
2015 – 2019	National Doctoral Fellowship, CONACYT, Mexico (4 years)
2013 – 2015	National Master’s Fellowship, CONACYT, Mexico (2 years)

Honors and distinctions

2021 – 2023	Candidate Researcher (Candidato a Investigador Nacional), National System of Researchers (SNII), Mexico
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Travel awards

2024	College of Science Travel Award, University of Idaho (US\$1,000) — SIAM Annual Meeting 2024
2023	Landahl Travel Award, Society for Mathematical Biology (US\$450) — SMB Annual Meeting 2023

TEACHING EXPERIENCE

Instructor of Record — MATH 483/4830: Foundations of Machine Learning Fall 2024, Fall 2025

Department of Mathematics and Statistical Science, University of Idaho

Sole instructor responsible for lectures, coding demonstrations, assignments, and grading; taught in English.

Fall 2024: 11 students (7 undergraduate, 2 M.Sc., 2 Ph.D.). Fall 2025: 12 undergraduate students.

Student evaluations, Fall 2025 (10 of 12 responding): mean 3.7/4.0 across 21 instructor and course items, above the department mean on 18 of 21.

Co-Instructor — Thermodynamics

Feb – Apr 2022

Center for Applied Physics and Advanced Technology (CFATA), UNAM, Juriquilla, Querétaro, Mexico

Teaching Assistant — Machine Learning

Feb – Jun 2021

Faculty of Sciences, UNAM, Mexico City, Mexico

MENTORING

Havilah Neujahr — Ph.D. co-advisor

Fall 2025 – present

Bioinformatics and Computational Biology Program, University of Idaho

Project (tentative): *AI-assisted modeling to untangle immune response mechanisms in co-infection*

PRESENTATIONS

Contributed talks

Jul 2026	ECMTB/SMB Joint Annual Meeting, Graz, Austria
Jun 2026	WNAR Annual Meeting, Pullman, WA, USA
Jun 2025	5th IFAC Workshop on Control of Systems Governed by PDEs, Beijing, China (<i>conference paper</i>)
Sep 2024	World Automation Congress (WAC 2024), Cancún, Mexico (<i>conference paper</i>)
Jul 2024	SIAM Annual Meeting (AN24), Spokane, WA, USA
Jun 2023	SMB Annual Meeting, Columbus, OH, USA
Jun 2021	2nd Annual Meeting of MexSIAM, online
Aug 2016	2nd International Workshop on Matter Out of Equilibrium, Guanajuato, Mexico

Posters

Jul 2019	Traffic and Granular Flow (TGF 2019), Pamplona, Spain
Jan 2019	XLVIII Winter Meeting on Statistical Physics, Puebla, Mexico
Jun 2018	30th International Conference on Science and Technology of Complex Fluids, San Luis Potosí, Mexico
Jan 2018	XLVII Winter Meeting on Statistical Physics, Puebla, Mexico
Jan 2016	XLV Winter Meeting on Statistical Physics, Taxco, Mexico

PROFESSIONAL SERVICE

Thesis committees

May 2026	M.Sc. in Mathematics, University of Idaho — Benjamin Whipple, <i>Modeling mechanisms of late-phase and spatial control of within-host influenza A virus infection</i>
Apr 2025	M.Sc. in Statistics, University of Idaho — Brett Flerchinger, <i>Multivariate mixed effects model on mouse infection</i>
Jul 2021	B.Sc. in Biomedical Physics (supervised professional practice), Faculty of Sciences, UNAM — Fernanda Ordóñez Jiménez, <i>Mathematical modeling to decipher multifactorial events in severe COVID-19 patients</i>

Journal referee (13 manuscripts)

PLOS Computational Biology (3), *Mathematical Biosciences and Engineering* (3), *Frontiers in Cellular and Infection Microbiology* (2), *Annual Reviews in Control* (2), *Frontiers in Microbiology* (1), *Journal of the Franklin Institute* (1), *Social Networks* (1).

Professional affiliations

2023 – present	Society for Mathematical Biology (SMB)
2024	Society for Industrial and Applied Mathematics (SIAM)

SOFTWARE AND TECHNICAL SKILLS

Research software

CrossLabFit · github.com/rblanco89/CrossLabFit

GPU-accelerated differential evolution for calibrating mechanistic models with qualitative constraints, integrating data across multiple laboratories (Python/CUDA; MIT license; accompanies J10).

Research code (MIT license):

- **covid-network** · github.com/rblanco89/covid-network — stochastic epidemic simulations on contact networks (C++; accompanies J7).
- **siloHopper_3D** · github.com/rblanco89/siloHopper_3D — GPU granular-flow simulations of silo-hopper discharge (CUDA; accompanies J2).

Technical skills

<i>Modeling</i>	ODE/PDE models, stochastic and agent-based models, epidemic network models, molecular dynamics, Monte Carlo, discrete element methods
<i>Inference and data</i>	Parameter estimation, model calibration and identifiability, machine learning, clustering, topological data analysis
<i>Programming</i>	C, CUDA, Python (Jupyter), Bash (advanced); C++, R, $\text{\LaTeX}$
<i>High-performance computing</i>	GPU computing; Linux clusters: RCDS (University of Idaho), Lemhi (Idaho National Laboratory), Kukulcan (Cinvestav Mérida), Abacus (Cinvestav); NPU servers with Docker (UNAM–Huawei project)
<i>Languages</i>	Spanish (native), English (professional working proficiency)

REFERENCES

Prof. Esteban A. Hernández-Vargas

Postdoctoral supervisor

Associate Professor, Dept. of Mathematics and Statistical Science

University of Idaho, USA

esteban@uidaho.edu

Prof. Tanya A. Miura

Collaborator

Professor and Chair, Department of Biological Sciences

University of Idaho, USA

tmiura@uidaho.edu

Prof. Gabriel Pérez-Ángel

Doctoral advisor

Professor, Department of Applied Physics

Cinvestav, Mérida, Mexico

gperez@cinvestav.mx

Prof. Timothy R. Johnson

Teaching reference

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